

12.1: Vectors:

A vector can be shown in 2 ways:

⇒ Magnitude - Dir^ ⇒ (r, θ)

⇒ Component Form ⇒ $a\underline{i} + b\underline{j}$ / $\begin{pmatrix} a \\ b \end{pmatrix}$

For 2-D vectors in 2D, the angle is measured from the positive x-axis.

The length (/modulus / magnitude) of a vector is similar to pythagoras. Notation: $|\vec{r}|$

$\vec{r} = a\underline{i} + b\underline{j} \Rightarrow |\vec{r}| = \sqrt{a^2 + b^2}$

Such as that $|l|$ where l is a line segment is:

$l = \begin{pmatrix} a \\ b \end{pmatrix} - \begin{pmatrix} c \\ d \end{pmatrix} \Rightarrow |l| = \sqrt{(a-c)^2 + (b-d)^2}$

$(r, \theta) \Rightarrow \begin{pmatrix} r\cos\theta \\ r\sin\theta \end{pmatrix} \Rightarrow (r\cos\theta)\underline{i} + (r\sin\theta)\underline{j}$

12.2: Working with Vectors:

The Position Vector of Point P is defined as $\vec{OP}$

It is the vector joining the origin to P.

For $\vec{AB}$, go $\vec{AO} + \vec{OB} = \vec{OB} - \vec{OA}$ $(\vec{OA} = -\vec{AO}) \Rightarrow \vec{AB} = \vec{OB} - \vec{OA}$ / b - a

A unit vector is a vector which has the same direction, but a magnitude of 1.

$r = a\underline{i} + b\underline{j}$

$\frac{\underline{r}}{|r|} = \hat{r} = \frac{a\underline{i}}{\sqrt{a^2+b^2}} + \frac{b\underline{j}}{\sqrt{a^2+b^2}}$

It has the same dir^ as it is only a scalar multiple of $\frac{1}{|r|}$!

Vectors are parallel when they share the same unit vector, or when $\mu\underline{a} = \lambda\underline{b}$. If $\mu = \lambda$, they are the same vector.

$\hat{a} = \pm\hat{b}$

Adding Vectors:

When adding vectors, individual components must be added. This will create a resultant vector.

Subtracting Vectors:

The same as adding the negative of a vector.